

Tezpur University
End Semester examination (Autumn-2025)
CS519: Computer Networks

Full marks=50

Time = 120 mins

(The figures in the right-hand margin indicate full marks for the questions)

1. **6+6=12**
Consider instances of TCP's-
 - a. TAHOE algorithm,
 - b. RENO algorithm,where the window size at the start of the slow-start phase is 1 MSS and the first slow-start threshold is set to 8 MSS. Assume TCP use over a lossy link i.e., congestion occurs after transmission at 10th RTT, Round Trip Time-Out expires at 15th transmission. Deduce the congestion window sizes at the 20th RTT for both the algorithms (in MSS).
2. **2+2+2+2=8**
Answer the followings:
 - a. Differentiate between an Autonomous System and an Area.
 - b. Use of local_pref and Multi-Exit Discriminator in BGP.
 - c. Enhancements in RIPv2 over RIPv1,
 - d. Name any two OSPF messages with their functionalities.
3. **5+1+1=7**
 - a. What are the transport layer protocols used for Telnet, real time multimedia, TFTP, DNS and email, respectively?
 - b. What is the use of queues in UDP?
 - c. What is a control block diagram used in UDP?
4. **4+4=8**
 - a. What are the characteristics of Gratuitous ARP?
 - b. Show valid TCP/IP headers with related source and destination addresses to encapsulate BOOTP packet and communicate from client to BOOTP server.
5. **2+2+2=6**
 - a. How does ICMP gateway redirect work?
 - b. Write any two valid code values with the reason statements for Type 3 ICMP.

- c. Which ICMP helps in detecting a firewall in the network? Mention the type value of the same ICMP message.

6.

3x3=9

Answer any three of the followings:

- a. Differentiate between mail transfer and mail access protocols. Name any two mail access protocols.
- b. Differentiate between persistent and non-persistent HTTP. How many RTTs may be required to establish a TLS connection with the server given DNS resolution is already done?
- c. What are the private IP address ranges?
- d. A company has a network address of 210.209.204.0/22. It wishes to have three subnets, one with 65 hosts and two with 35 hosts each. Write down a feasible set of subnet address/subnet mask pairs for minimal address space assignments?
- e. Briefly explain the difference between symmetric-key cryptography and asymmetric-key cryptography. Give one common algorithm for each type.
- f. What are confidentiality and integrity in the context of cryptography? Which one is typically addressed by encryption, and which one by hashing (or Message Authentication Codes)?
